

SIMATS ENGINEERING
ASSESSMENT TOOL 3: Application Based Questions

COURSE NAME: Cloud Computing and Big Data Analytics **COURSE CODE:** CSA15

COURSE OUTCOME COVERED

CO2: Analyze virtualization architectures to identify performance and security issues in cloud computing environments. (BL3)

Assessment Tool Weightage: 10%

Dave's Taxonomy: Manipulation (Level 2)
Question Count: 5

Total Marks: 25

Code of Conduct

I M.V.Rakesh Reddy/192524164(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: M.V.Rakesh Reddy_____

Assessment Tasks

Application Based Questions

- 1.A company experiences unpredictable traffic spikes during seasonal sales. Explain how cloud auto-scaling and load balancing can handle this demand efficiently. How does this approach reduce downtime and improve user experience? What role do monitoring tools play in this setup? Relate your answer to a real-world e-commerce scenario.
- 2.An organization wants to store large volumes of backup data securely in the cloud.
 - Discuss how cloud storage solutions ensure durability and data redundancy.
 - How can encryption and access control protect sensitive backup data?
 - Explain the cost advantages compared to traditional storage systems.
 - Provide an example of a disaster recovery use case.
- 3.A mobile app company plans to deploy its application globally. Explain how content delivery networks (CDNs) improve performance and reduce latency. How do edge servers enhance user experience across regions? Discuss how cloud providers support global availability. Relate this to a real-time streaming or gaming application.

4.A financial institution requires strict compliance and data security in cloud usage.

Explain how cloud providers support regulatory compliance and auditing. What role do logging, monitoring, and encryption play in security? How can the company ensure data sovereignty requirements are met? Give an example involving sensitive financial transactions.

5.A development team wants to adopt containerization for application deployment.

Explain how containers differ from virtual machines in terms of performance and resource usage. How does orchestration (like Kubernetes) simplify deployment and scaling? Discuss portability benefits across different cloud environments.

Provide an example of a microservices-based application.

Application Based Questions Rubric

Criteria Relevance to Application	Weightage 25%	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
		Response directly addresses the given use case with strong relevance	Mostly relevant response	Partially relevant	Weak relevance
Use of Cloud Concepts	25%	Applies appropriate concepts accurately	Mostly accurate application	Basic application only	Incorrect concept usage
Practical Insight	20%	Shows strong real-world understanding	Shows reasonable practical understanding	Limited practical insight	No practical insight
Completeness	15%	Response covers all required aspects	Covers most aspects	Covers only basic aspects	Incomplete response
Clarity	15%	Clear and concise answer	Generally clear	Some confusion in expression	Poor clarity

1. Traffic Spikes & E-Commerce (Auto-Scaling & Load Balancing)

Task Description: Explain how cloud auto-scaling and load balancing handle predictable traffic spikes during seasonal sales, reduce downtime, improve user experience, and utilize monitoring tools based on an e-commerce scenario[cite: 2].

Handling Demand Efficiently:

- Cloud auto-scaling dynamically adjusts computing capacity by automatically provisioning extra virtual machine instances during traffic surges and scaling them down when traffic subsides[cite: 2].
- A load balancer acts as a traffic controller, distributing incoming user requests evenly across multiple backend server instances to handle high demand without overwhelming any single server[cite: 2].

Downtime Reduction & User Experience:

- Intelligent load management and automated scaling prevent server crashes, HTTP response errors, and slow page loading, ensuring uninterrupted checkout workflows and consistent responsiveness[cite: 2].

Role of Monitoring Tools:

- Real-time monitoring tools track infrastructure health metrics continuously and trigger automated scaling policies instantly when preset traffic thresholds are exceeded[cite: 2].

Real-World E-Commerce Scenario:

- During a major seasonal flash sale (such as Black Friday or Diwali), a platform like Amazon experiences a sudden surge in user logins and transactions[cite: 2].
- Auto-scaling adds compute instances instantly to absorb the load, while the load balancer splits traffic evenly to guarantee zero downtime[cite: 2].

2. Secure Cloud Backup Storage & Disaster Recovery

Task Description: Discuss how cloud storage solutions ensure durability, redundancy, encryption, access control, and cost advantages compared to traditional systems, supported by a disaster recovery use case[cite: 2].

Durability and Data Redundancy:

- Enterprise cloud storage solutions achieve extreme data durability by automatically replicating backup objects synchronously across multiple availability zones and separate physical data centers[cite: 2].

Encryption and Access Control:

- Data is heavily secured using encryption at rest and in transit via secure keys, ensuring information remains unreadable if intercepted[cite: 2].
- Robust Identity and Access Management (IAM) policies and multi-factor authentication prevent unauthorized entry and protect sensitive backup archives[cite: 2].

Cost Advantages:

- Unlike traditional capital-intensive on-premises physical SAN/NAS or tape systems, cloud storage operates on an economical pay-as-you-go operational expenditure (OpEx) model that eliminates upfront hardware costs[cite: 2].

Disaster Recovery Use Case:

- Following a severe ransomware attack that cripples primary local servers, an organization can swiftly restore clean virtual machine images and corporate databases from secure cloud snapshots to resume operations[cite: 2].

3. Global App Deployment via CDNs and Edge Computing

Task Description: Explain how content delivery networks (CDNs) and edge servers improve performance, reduce latency, and support global availability for streaming or gaming applications[cite: 2].

Improving Performance and Reducing Latency:

- Content Delivery Networks (CDNs) store static and dynamic content across globally distributed edge servers situated close to end-users, bypassing high-latency long-distance data transfers[cite: 2].

Role of Edge Servers:

- Edge nodes process requests locally, perform SSL termination, and handle caching, significantly lowering origin server workloads and speeding up response times[cite: 2].

Global Availability Support:

- Worldwide cloud provider backbones with multi-region points of presence (PoPs) ensure high availability, resilient DNS routing, and global traffic management[cite: 2].

Real-Time Streaming / Gaming Scenario:

- In multiplayer gaming or live video streaming platforms, high latency causes lag and buffering[cite: 2].
- CDNs cache media manifest files and game maps near gamers worldwide, ensuring ultra-low latency playback and a seamless user experience[cite: 2].

4. Compliance, Auditing, and Data Sovereignty in Finance

Task Description: Explain how cloud providers support regulatory compliance, auditing, security logging, encryption, and data sovereignty requirements using a sensitive financial transaction example[cite: 2].

Regulatory Compliance & Auditing:

- Cloud providers adhere strictly to global security frameworks (such as PCI-DSS, GDPR, HIPAA) and offer automated auditing dashboards and structured shared responsibility matrices[cite: 2].

Role of Logging, Monitoring, and Encryption:

- Immutable system logs track all user actions and API interactions to detect intrusions or anomalies instantly[cite: 2].
- Encryption keeps financial records and account details completely unreadable to unauthorized entities both in storage and during transit[cite: 2].

Data Sovereignty Requirements:

- Organizations can enforce geographic data residency limits explicitly to ensure regional records stay within compliant data centers that align with local legal frameworks[cite: 2].

Financial Transaction Example:

- During an international online wire transfer through a banking application, transaction details are encrypted, processed in a compliance-audited regional zone, and logged permanently for auditing[cite: 2].

5. Containerization vs. Virtual Machines and Orchestration

Task Description: Explain the differences between containers and virtual machines regarding performance and resource usage, how orchestration simplifies scaling, portability benefits, and a microservices use case[cite: 2].

Performance and Resource Differences:

- Virtual machines run a heavy hypervisor with an independent guest operating system per instance, leading to heavy resource consumption[cite: 2].
- Containers share the host operating system's kernel, making them remarkably lightweight, fast to spin up, and highly resource-efficient[cite: 2].

Role of Orchestration (Kubernetes):

- Kubernetes automates container deployment, scaling, network load distribution, and self-healing across cluster environments[cite: 2].

Portability Benefits:

- Containers bundle applications alongside all dependencies, ensuring environment-agnostic execution across developer machines, private servers, and multi-cloud platforms without drift[cite: 2].

Microservices-Based Application Example:

- A food delivery platform splits functionalities like user login, restaurant search, and payment processing into separate microservices[cite: 2].
- Each service runs inside an independent container scaled dynamically via Kubernetes to handle traffic peaks independently[cite: 2].